

Perovskite Photovoltaics in January-June 2026: The Certified Frontier From Cell to Module

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Abstract

Over an evaluation period spanning 6 months, the metal halide perovskite literature expanded by 3328 indexed publications, of which 884 were examined closely to assess physical claims, interfacial mechanisms, and stability validation across distinct device architectures. Across single junction devices, the certified power conversion efficiency (PCE) ceiling advanced to 27.6%, yet mechanistic investigations diverged over the operational function of low-dimensional capping heterostructures: while certain studies concluded that wide-bandgap surface caps suppress non-radiative recombination and impede halide migration without hindering transport, competing evidence demonstrated that disordered organic spacer phases create severe charge-transport barriers, accumulate photogenerated carriers under illumination, and penalize fill factor (FF). In all-perovskite tandems, performance ceilings remained strictly governed by the chemical vulnerability of narrow-bandgap (NBG) mixed tin-lead sub-cells; although chemical hardness engineering and nucleation control synchronized crystallization to suppress localized tin-rich defect segregation, physical evidence that initial crystallization control arrests autonomous divalent-to-tetravalent tin oxidation cascades during continuous operation remains thin and contested. Monolithic hybrid tandems paired with silicon or chalcogenide bottom absorbers established the issue-wide ceiling of 34.85% certified PCE, but exposed an acute mechanistic dispute regarding buried hole-selective layer (HSL) contacts: conflicting investigations dispute whether dense intermolecular pi-pi stacking and strict crystalline order within self-assembled monolayers (SAMs) are mandatory to align interfacial energy levels and eliminate pinholes, or whether crystalline domains induce nanoscale grain boundaries and thermal dewetting that require sterically hindered, amorphous molecular geometries to preserve conformal passivation. Translating these absorbers into perovskite solar modules (PSMs) incurred a steep area penalty, driving certified PCE down to 26.8% at the largest reported aperture area of 655.2 cm², with reports conflicting over whether this scaling penalty is dominated by non-radiative recombination at laser scribing margins, convective drying instabilities generating coffee-ring solute segregation, or heterogeneous out-of-plane residual lattice strain across the bulk film. Operational stability verification remains critically sparse and fragmented across all architectures: only 39 publications designated a standardized consensus protocol from the International Summit on Organic Photovoltaic Stability (ISOS), and only 39 studies reported an

operational lifetime metric, with the longest duration reaching 4100 h under continuous illumination. Reporting rigor varies widely across device families, as 52.1% of published abstracts state an efficiency value, yet only 8.5% report independent certification, representing just 114 certified works across the surveyed corpus. This issue concludes that progress in perovskite photovoltaics has decoupled from uncertified efficiency metrics, shifting fundamentally toward unraveling coupled ionic-electronic degradation under operating electrical and thermal stress; resolving buried contact degradation between out-of-plane molecular desorption and lateral field-induced dewetting requires operando bias-dependent sum-frequency generation and conductive atomic force microscopy, while settling open-circuit voltage (Voc) losses demands operando drive-level capacitance profiling (DLCP) coupled with depth-resolved photoluminescence quantum yield (PLQY) mapping to distinguish true defect passivation from electrostatic band realignment.

Keywords: perovskite photovoltaics; buried interface; defect passivation; wide-bandgap perovskite; tandem solar cells; operational stability

1. Introduction

Investigations into operational degradation in halide perovskites yield incompatible conclusions regarding the primary cause of open-circuit voltage loss under bias and illumination. One body of work attributes performance decay to electric-field-driven migration of halide ions and subsequent barrier accumulation at selective extraction contacts, whereas an opposing set of studies identifies photo-induced bulk phase segregation and deep trap formation within the absorber as the governing mechanism. Correlative operando nanoscale chemical profiling paired with depth-resolved transient photoluminescence under operational bias is required to resolve whether performance decay originates at transport boundaries or in the bulk lattice. This controversy exemplifies how the frontier of perovskite photovoltaics has shifted away from the simple pursuit of initial power conversion efficiency (PCE). The central challenge now centers on operational endurance: mitigating coupled ionic-electronic degradation, maintaining interfacial band alignments, and arresting defect propagation under continuous operating conditions.

Structuring the contemporary literature by device family provides the only rigorous framework for evaluation, as each architecture operates under distinct physical boundary conditions and occupies a different tier of experimental maturity. The field is divided into four distinct categories: the single-junction frontier and its remaining losses, all-perovskite tandems: narrow-bandgap stability as the limit, hybrid tandems: silicon, cigs and organic partners, and modules and the cost of area. Single-junction devices confront intrinsic non-radiative surface recombination and remaining bulk voltage deficits. In all-perovskite tandems, performance is restricted by the oxidation susceptibility and high defect density of tin-lead narrow-bandgap subcells. Hybrid tandems pairing perovskites with silicon, copper indium gallium selenide (CIGS), or organic absorbers face current-matching restrictions, parasitic optical absorption, and thermal processing incompatibilities. Finally, module architectures incur substantial geometric fill factor losses, non-uniform deposition across expanded substrates, and resistive dissipation through transparent conductive oxides.

A critical audit of the published literature establishes a sharp contrast between unverified hero metrics and authenticated performance. Across the six-month survey, 3328 indexed works met the scope gate across six months and 884 were read closely. In the single-junction category, covering 692 papers, the best certified PCE was 27.6%, but only 63 of them reported any certified value, with a largest area of 4 cm² and a longest time to 80% initial performance (T80) of 4100 h. For all-perovskite

tandems across 60 papers, the best certified PCE reached 30.3%, 15 of them reported any certified value, the largest area was 64.64 cm^2 , and the longest T80 was 1600 h. In the hybrid tandem literature comprising 77 papers on perovskite/silicon and other hybrid tandems, the best certified PCE reached 33.6%, 19 of them reported any certified value, the largest area was 16 cm^2 , and the longest T80 was 1000 h. For modules, represented across 43 papers, the best certified PCE was 26.8%, 10 of them reported any certified value, the largest area was 900 cm^2 , and the longest T80 was 1050 h. Across each category, this review separates what was verified from what was asserted.

2. What January-June 2026 Verified, by Device Family

Incompatible physical interpretations persist regarding whether open-circuit voltage (Voc) losses and operational decay in metal halide perovskites originate primarily from interfacial band misalignment or from unpassivated chemical defects. Proponents of electrostatic interface engineering argue that tuning interfacial dipole moments and work functions across hole-transport layer contacts eliminates non-radiative recombination and governs extraction kinetics [1,2]. Conversely, coordination chemistry studies conclude that carrier losses are dominated by uncoordinated metal sites and localized phase segregation, which require multidentate Lewis-base chelators or redox mediators to stabilize the lattice [3,4]. Resolving this conflict requires operando bias-dependent drive-level capacitance profiling paired with depth-resolved photoluminescence quantum yield mapping under coupled electrical and thermal stress, isolating electrostatic potential steps from true defect deactivation. Interpreting performance metrics across different device categories requires establishing that a single-junction cell and a two-junction tandem are bounded by different physics, so a higher tandem number is not a better cell. The leading certified values belong to fundamentally different device architectures that divide the solar spectrum across distinct thermodynamic regimes rather than indicating superior absorber quality. In addition, a value certified on a square-centimetre cell and one certified at module area are different kinds of evidence, separated by interconnect resistance, geometric dead area, and spatial deposition non-uniformities.

For single-junction devices, the literature comprised 692 papers, but only 63 of them reported any certified value, demonstrating that uncertified laboratory measurements represent the vast majority of published claims. The certified power conversion efficiency (PCE) ceiling moved upward over the half year rather than repeating a static value. Certified single-junction PCE began at 27.1% in January [5], registered 26.9% in February [3], 26.78% in March [6], and 27.03% in April [7]. The ceiling reached a best certified 27.6% in May through entropy-regulating molecular locking of the crystal lattice [8], before recording 27.28% in June [9]. In contrast to efficiency gains, verified scaling and operational durability remained tightly constrained: the largest area verified was 4 cm^2 [10,11], and the longest time to 80% of initial performance (T80) reached 4100 h [12]. These values verify that high single-junction performance remains concentrated in small-area cells without standardized durability tracking.

Tandem architectures similarly verified moving performance ceilings, driven by distinct spectral-splitting mechanics. All-perovskite tandems were examined in 60 papers, with 15 of them reporting any certified value. Their certified ceiling moved across the period, recording 29.2% in January [13], 20.02% in February [14], 27.7% in March [15], and advancing to a best certified 30.3% in April through chemical hardness synchronization [16], followed by 29.44% in May [17] and 28.79% in June [18]. The largest area verified for all-perovskite tandems was 64.64 cm^2 [15], while the longest T80 reached 1600 h [19], reflecting the persistent vulnerability of narrow-bandgap mixed tin-lead subcells

to tin oxidation. Perovskite/silicon and other hybrid tandems encompassed 77 papers, with 19 of them reporting any certified value. Their certified ceiling moved upward from 28.25% in January [20] to 32.46% in February [21], 31.93% in March [22], 32.81% in April [23], 33.5% in May [24], and reached a best certified 33.6% in June using an indium-free contact scheme [25]. However, the largest verified area for hybrid tandems was 16 cm² [26] and the longest T80 reached 1000 h [27], demonstrating that hybrid records remain confined to small test apertures.

The module family totaled 43 papers, with 10 of them reporting any certified value, confirming that independent verification is exceptionally rare at large area. For monolithic perovskite modules, the certified ceiling repeated within a narrow efficiency band rather than establishing an upward trajectory. Certified monthly values tracked 25.62% in January [28], 26.75% in February [29], None in March, 25.55% in April [30], and 26.64% in May [31], establishing a best certified module efficiency of 26.8% [32]. Although June recorded 32.95% [33], that certification belonged to an integrated perovskite/silicon tandem cell architecture reported within a module study, not a monolithic perovskite module aperture, confirming that architectural categories cannot be combined into a single progression. The largest area verified across modules was 900 cm² [34], and the longest T80 reached 1050 h [35]. These metrics verify that module development remains governed by interconnect series resistance and film non-uniformity across large substrate areas rather than absorber efficiency limits.

Table 1. State of the art by device family, January-June 2026

Device family	Papers	Certified (n)	Certified (%)	Best certified PCE	Best self-reported PCE	Largest area	Longest T80
Single junction	692	63	9.1%	27.6%	28.01%	4 cm2	4100 h
All-perovskite tandem	60	15	25%	30.3%	30.3%	64.64 cm2	1600 h
Hybrid tandem	77	19	24.7%	33.6%	34.02%	16 cm2	1000 h
Module	43	10	23.3%	26.8%	27.12%	900 cm2	1050 h

Device families are abbreviated in the first column: “Hybrid tandem” is a perovskite absorber paired with a non-perovskite partner, silicon in most cases and otherwise an organic, chalcopyrite, telluride or kesterite cell. Papers are depth-tier records assigned to exactly one family by the device named in the quotation carrying each number, not by the paper’s title. Best-certified values exclude any value whose own quotation names a different device family, so a tandem-cell record reported inside a module study is not counted as a module result. Simulation and review papers are excluded from every measured value, as are efficiencies measured under indoor or low-light illumination, which are not comparable to one-sun values. 47 tandem papers name no partner and are grouped with all-perovskite tandems; they are reported separately in the Supplementary Information. An em-rule means no value was reported, which is not the same as a reported zero.

Table 2. Best independently certified device in each family, with provenance

Device family	Certified PCE	Aperture area	Architecture	Month	Venue
Single junction	27.6%	–	unknown	May	Science
All-perovskite tandem	30.3%	–	tandem_2T	Apr	Nature Nanotechnology
Hybrid tandem	33.6%	1 cm ²	tandem_2T	Jun	Science
Module	26.8%	655.2 cm ²	p-i-n	Jun	Advanced Materials

Each row resolves to exactly one extraction record, cited in the column at the end of the corresponding row in the Supplementary Information, and every value appears verbatim in the cited paper’s own abstract. Aperture area is shown only where the same paper reported it; an em-rule means the area was not stated alongside the certified value.

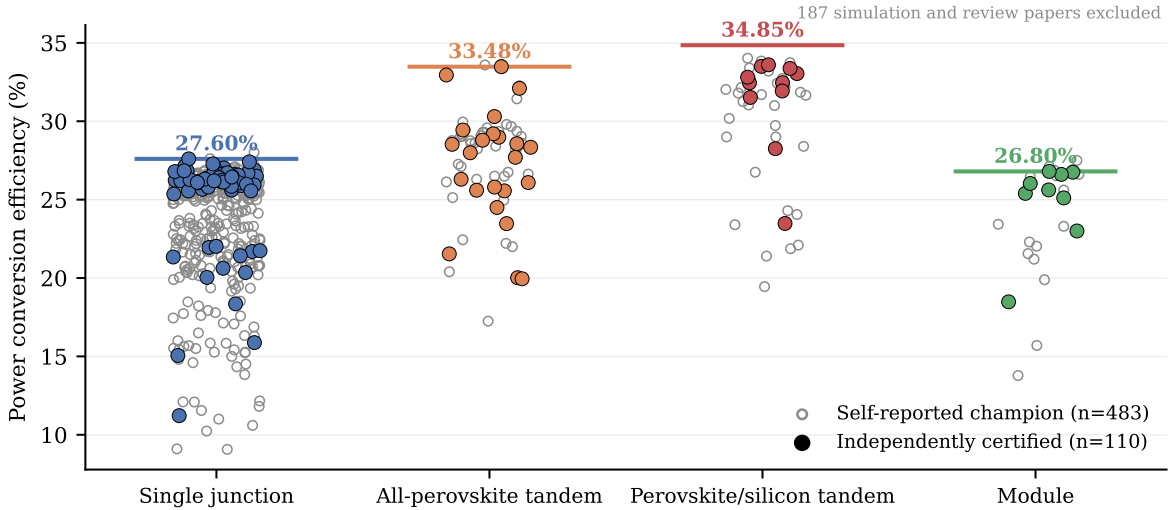


Figure 1. The certified frontier in January-June 2026. Filled markers are independently certified efficiencies, open markers are self-reported champion values, grouped by device family. Horizontal bars mark the highest certified value in each family.

3. The Single-Junction Frontier and Its Remaining Losses

The single-junction frontier and its remaining losses reveal a sharp physical contradiction dividing recent efforts to eliminate non-radiative recombination at the upper perovskite interface in perovskite solar cells (PSCs): whether low-dimensional capping layers suppress interfacial defect states without electronic cost, or establish resistive transport barriers that choke charge extraction. One body of work concludes that low-dimensional surface reconstruction passivates undercoordinated halide vacancies and blocks ion migration without penalizing collection. Dimensional control of low-dimensional perovskite hybrids achieved a certified power conversion efficiency (PCE) of 27.02% (27.21% champion, tested under an unspecified International Summit on Organic Photovoltaic Stability protocol, ISOS-L-2I) [36], multivalent ligand-regulated dimensional engineering

delivered a certified 25.4% PCE (25.4% champion on 1.1 cm^2) [37], and in situ formation of a guanidinium derivative two-dimensional/three-dimensional (2D/3D) heterostructure yielded a certified 26.2% PCE (26.49% champion) [38]. Conversely, bulky organic spacer cations in two-dimensional (2D) capping networks introduce severe electronic insulation that impedes out-of-plane transport [39], while solution-deposited 2D layers form disordered quantum-well phase distributions that restrict extraction and depress fill factor (FF). To bypass this transport resistance, liquid-derived, solvent-free vapor-mediated dimensional reconstruction eliminated disordered insulating phases, achieving a certified 26.15% PCE (26.71% champion on 1 cm^2) with a high fill factor [40]. This conflict exposes an unverified premise: whether solution-deposited 2D capping acts as benign passivation or as a resistive bottleneck masked by small test areas. Settling this disagreement requires operando cross-sectional conductive atomic force microscopy synchronized with bias-dependent transient absorption spectroscopy to map localized potential drops and carrier transit delays across the heterojunction under working illumination.

Across 692 single-junction publications evaluated over six months, non-radiative losses concentrate at interfaces (176 papers) and defect sites (196 papers). The buried interface between the transparent conducting oxide and absorber remains a primary locus of non-radiative voltage loss and mechanical debonding. Self-assembled monolayers (SAMs) anchored to nickel oxide (NiO_x) or indium tin oxide frequently suffer from island agglomeration, pinhole shunts, and labile coordinate bonds that dissociate under thermal stress. Strengthening substrate anchoring of polymeric self-assembled monolayers suppresses interfacial mechanical delamination, delivering a certified 26.5% PCE (26.9% champion on 1 cm^2) [41]. Hydroxyl interfacial engineering for self-assembled monolayer anchoring on NiO_x reduces interfacial trap densities to yield a certified 26.2% PCE (26.6% champion on 0.0524 cm^2) [42]. Modifying assembly kinetics with amino acid hydrochloride-mediated self-assembled monolayers reached a certified 26.33% PCE (26.87% champion on 0.058 cm^2) [43], while homogenizing interfacial assembly via indole-mediated binary monolayers eliminated local potential fluctuations, yielding a certified 26.57% PCE (26.95% champion on 0.0717 cm^2) [44]. Dual-site passivation within rationally designed self-assembled monolayers coordinates undercoordinated lead and halide vacancies simultaneously, delivering a certified 26.13% PCE (26.58% champion) [45]. To decouple transport from recombination, hole-selective monolayer molecules with spatially separated carrier orbitals and twisted pi-skeletons reached a certified 25.11% PCE (25.61% champion) [46]. However, electrostatic shifts alone cannot eliminate chemical vacancies: interfacial dipole engineering in both inverted (p-i-n) and normal (n-i-p) architectures attained a certified 26.23% PCE (26.18% champion) [47], yet uncoordinated precursor species at the buried contact require competitive-binding buried interlayers to maintain reproducible crystallization, achieving a certified 25.8% PCE (26.3% champion on 0.07 cm^2) [48]. Additionally, heterogeneous nucleation templates on the buried interface enable antisolvent-free deposition on flexible substrates, yielding a certified 26.64% PCE (26.89% champion) [49], while stabilizing sputtered NiO_x via in situ dissociative adsorption passivation curtails contact pinhole shunts to yield a certified 26.07% PCE (26.31% champion) [50].

Within the perovskite bulk, non-radiative decay reflects the competition between crystallization kinetics and structural defect formation, addressed by 143 papers on composition. Thermal annealing typically induces volatile organic cation escape and leaves tensile lattice strain from thermal expansion mismatch, accelerating ion migration under bias. Over the six-month survey, research shifted away from static additive screening toward mechanical and dynamic lattice regulation, reflecting the recognition that chemical passivation alone has plateaued near the 26.22% median certified efficiency. Molecular press annealing counteracts tensile distortion by applying physical compression

during thermal treatment, compressing lattice strain and preventing volatile loss to yield a certified 26.5% PCE (26.6% champion) [51]. Contact-triggered molecular interactions direct crystal growth to achieve a certified 25.61% PCE (26.25% champion) [52], whereas mechanochemically reinforced dual-dynamic covalent seeding guides grain orientation and retards phase segregation, producing a certified 26.69% PCE (26.95% champion under ISOS-L-2I testing) [53]. Chemical additives targeting point defects also suppress non-radiative traps: fused-ring electron acceptors regulate crystallization and passivate deep electronic states, yielding a certified 26.35% PCE (26.66% champion) [54], while multifunctional multi-interaction additives achieve a certified 25.66% PCE (26.1% champion) by coordinating undercoordinated lead ions [55]. In wide-bandgap 1.68-electron-volt (1.68-eV) absorbers, grain boundary defect density was reduced through molten phosphonoalkanoic acid-driven passivation in hybrid evaporated/blade-coated deposition, delivering a certified 21.95% PCE (22.53% champion) [56]. Common chemical additives also homogenized optoelectronic properties across the film, yielding a certified 26.4% PCE on a 0.05 cm² area [57]. To bypass solvent-induced inhomogeneities, crystal-facet-directed all-vacuum thermal deposition yielded a certified 18.35% PCE (19.3% champion on 0.25 cm² under ISOS-L-2 operational tracking) [58]. Polymer-mediated synthesis achieved scalable air-processed deposition, yielding a certified 26.22% PCE (26.5% champion on 0.06 cm²) [59], matching the single-junction median certified efficiency of 26.22%. At the compositional apex of bulk optimization, pure formamidinium lead triiodide (FAPbI₃) single-junction architectures achieved the fleet-wide highest certified performance, stating that “We achieved a certified power conversion efficiency (PCE) of 27.6% in FAPbI₃-based PSCs” [8], although the authors reported no device aperture area.

Carrier collection losses at the upper interface between the perovskite absorber and charge transport layers represent an acute thermodynamic bottleneck, combining non-radiative Shockley-Read-Hall recombination with interfacial chemical degradation. Sputtered metal contacts and electron transport layers (ETLs) induce atomic interdiffusion and chemical reduction of lead halides. Depositing an ultrathin hafnium oxide interface stabilization layer between the fullerene electron transport layer and metal contact forms an impermeable barrier against volatile iodide out-diffusion and metal atom ingress, achieving a certified 26.6% PCE (27.1% champion) [60]. Molecular confinement engineering at the upper interface improves physical contact and blocks ion migration pathways, providing a certified 26.79% PCE (26.84% champion under ISOS-L-1 continuous illumination) [61]. Chemical coordination at this interface is stabilized by bifunctional donor-pi-acceptor (D-pi-A) ligands that direct self-organized interface passivation, achieving a certified 26.37% PCE (26.56% champion under ISOS-L-2 testing) [62], while spatial molecular engineering of hole semiconductors yielded a certified 26.24% PCE (26.6% champion under ISOS-L-1) [63]. For the hole transport layer in inverted and normal architectures, designing spiro-buckybowl-structured hole-transporting materials optimizes interfacial energy alignment and moisture resistance to attain a certified 25.36% PCE (25.54% champion under ISOS-L-2) [64]. Conjugated amine ligand-induced interlayers reinforce mechanical durability and electrical conduction, delivering a certified 26.29% PCE (26.33% champion under ISOS-L-3 damp-heat conditions) [65], and replacing conventional dopants with lithium-based deep eutectics in 2,2',7,7'-tetrakis[N,N-di(4-methoxyphenyl)amino]-9,9'-spirobifluorene (spiro-OMeTAD) suppresses dopant aggregation and ion redistribution, yielding a certified 26.37% PCE (26.53% champion) [66]. Photochemical degradation was mitigated by deuteration-functionalized self-assembled monolayers that resist carbon-hydrogen bond cleavage under ultraviolet (UV) irradiation, yielding a certified 25.9% PCE (26.34% champion) [67]. Uncertified interface treatments also reported high values: interfacial bridging engineering delivered an uncertified 26.87% champion PCE on a 0.045 cm² area

with an operational lifetime to 80% of initial performance (T80) of 500 h [68], and dip coating of passivation layers on formamidinium-cesium absorbers achieved an uncertified 22.5% champion PCE on a 0.105 cm² area with 1200 h T80 stability [69]. Polar charge extraction was also assisted by ferroelectric heterojunctions, which establish internal polarization to boost the open-circuit voltage-fill factor product, attaining a certified 26.07% PCE (26.62% champion) [70].

Beyond static defects, degradation mechanisms (investigated in 92 stability-focused studies) originate from dynamic lattice fatigue, operational temperature cycling, and defect propagation under illumination. Dynamic light-induced isomerization of photoswitchable molecular isomers repairs grain boundary bonds under continuous illumination, achieving a certified 26.9% PCE (27.2% champion) [3]. Positional methylation isomer additives alter molecular packing and defect migration barriers, yielding a certified 26.78% PCE (26.84% champion under ISOS-L-2 testing) [6]. Regulating lattice strain via molecular buffering dissipates thermomechanical stress under cryogenic cycling, yielding a certified 25.94% PCE and establishing the single-junction family’s highest self-reported PCE of 28.01% under ISOS testing [71]. In lead-free tin-based perovskites, rapid oxidation of tin from Sn²⁺ to Sn⁴⁺ introduces severe p-type self-doping and defect traps. Multidentate chelation by imine-linked covalent organic frameworks (COFs) suppressed tin oxidation and passivated coordination vacancies to yield a certified 15.88% PCE (16.29% champion) [72], while multifunctional reducing molecular dopants delivered a certified 15.06% PCE (16.0% champion) [73]. In mixed tin-lead cells, multifunctional MXenes integrated into NiO_x hole transport layers retarded tin oxidation and reduced energy offset losses, achieving a certified 20.04% PCE (22.5% champion) with a 600 h T80 lifetime [74]. In all-inorganic cesium lead triiodide (CsPbI₃), avoiding the phase transition to the non-photoactive yellow phase remains challenging: three-in-one self-assembled molecules achieved a certified 20.62% PCE (21.1% champion on 1 cm²) by passivating halide vacancies and optimizing energy alignment [75], and moisture-responsive crystallization enabled CsPbI₃ fabrication under high humidity to deliver a certified 22.02% PCE (22.6% champion) [76]. In low-dimensional perovskites, synergistic passivation delivered a certified 21.69% PCE (22.76% champion) in quasi-two-dimensional (quasi-2D) cells [77]. Extending beyond pure halides, molecular lead halide perovskite layers bridging silver bismuth sulfide (AgBiS₂) nanocrystals achieved a certified 11.22% PCE on a 0.022 cm² aperture [78]. Interparticle and interfacial electrostatic interactions maintained efficiency upon scaling, delivering a certified 26.03% PCE (26.47% champion on 0.0727 cm²) [79].

Rigorous evaluation reveals persistent reporting deficiencies across the single-junction literature. Across 692 evaluated papers (with 161 excluded as simulations or reviews), only 9.1% (63 papers) reported an independently certified efficiency, with a median certified PCE of 26.22% and a certified peak of 27.6% [8], while self-reported claims reached 28.01% [71]. Device aperture area was explicitly reported in only 6.6% of papers, with the largest reaching 4 cm², while the vast majority relied on sub-0.1 cm² test structures that minimize series resistance and mask lateral spatial inhomogeneities. Long-term operational stability was quantified via T80 in only 3.9% of studies (reaching a maximum of 4100 h), and merely 25 papers (3.6%) adhered to standardized protocols specified by the International Summit on Organic Photovoltaic Stability. In contrast, all-perovskite tandems achieved a certified 30.3% PCE (with 15 of 60 papers reporting certified values, a largest area of 64.64 cm², and a longest T80 of 1600 h), perovskite/silicon and hybrid tandems reached a certified 33.6% PCE (with 19 of 77 papers reporting certified values, a largest area of 16 cm², and a longest T80 of 1000 h), and perovskite modules attained a certified 26.8% PCE (with 10 of 43 papers reporting certified values, a largest area of 900 cm², and a longest T80 of 1050 h). Additionally, indoor photovoltaics must be evaluated separately from standard one-sun metrics because the narrow, low-flux indoor emission

spectrum permits high conversion percentages at drastically reduced absolute power densities. Wide-bandgap perovskite indoor photovoltaics achieved an indoor power conversion efficiency of 44.36% (producing an absolute power output of $127.94 \mu\text{W cm}^{-2}$ with an open-circuit voltage of 1.091 V) under indoor illumination [80], fully evaporated devices delivered indoor efficiencies of 41.60% at 900 lux and 41.22% at 300 lux illumination [81], a single-junction cell achieved an efficiency of 16.36% under 1000 lx illumination with operational stability exceeding 3000 h under nitrogen [82], and a stripped-down hybrid structure achieved an efficiency of 12.53% under 1,000 lux [83]. Several studies conflated illumination regimes by reporting indoor and one-sun efficiencies together in a single sentence: an efficiency of 35.54% under indoor light-emitting diode (LED) illumination and 20.28% under standard one sun conditions were reported together [84], an efficiency of 22.27% under air mass 1.5 global (AM 1.5G) illumination was reported together with indoor conditions [85], and an efficiency of 18.40% under 1 sun was reported together with 34.54% under indoor LED illumination (1000 lux) [86].

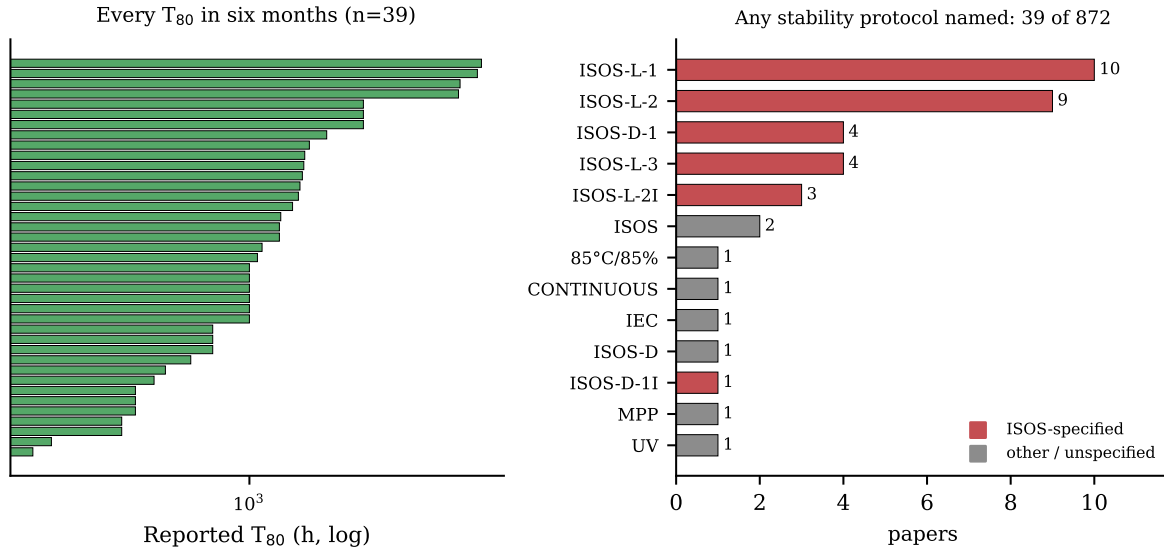


Figure 2. The month’s operational-stability evidence in full. Left, every reported time to 80% of initial performance. Right, the stability protocol labels used.

4. All-Perovskite Tandems: Narrow-Bandgap Stability as the Limit

Investigations over the past six months divide over whether narrow-bandgap (NBG) mixed tin-lead (Sn-Pb) sub-cell degradation is an extrinsic artefact of crystallization kinetics or an unavoidable electrochemical consequence of operating redox chemistry. One school of thought attributes Sn-Pb instability to differential crystallization kinetics, wherein tin iodide species nucleate and precipitate ahead of lead counterparts, generating Sn-rich localized phases that host high densities of tin vacancies and metallic tin [16,87]. Synchronizing crystallization rates through chemical hardness engineering achieved a certified power conversion efficiency (PCE) of 30.3% (champion 30.3% in an unspecified device area) [16], and suppressing Sn-rich phase precipitation enabled a certified PCE of 28.53% (champion 28.77%) [87]. Hybrid seeding-template modulation [88] (champion 29.18%, 500 h T₈₀, representing the time to 80% of initial efficiency) and hard Lewis base coordination [89] (champion 29.44%, 1080 h T₈₀) similarly operate on the premise that arresting disordered nucleation eliminates the primary initiation sites for chemical degradation. In contrast, competing evidence

demonstrates that degradation proceeds through autonomous redox cascades that operate independently of initial film morphology [18,90]. Reverse-bias operational stresses trigger self-propagating redox cascades between Sn^{2+} and oxidized halide species, driving structural breakdown even in dense, uniform absorbers [90] (champion 29.03%). In parallel, cyclic iodine redox reactions at the buried contact continuously regenerate triiodide that oxidizes Sn^{2+} to Sn^{4+} [18] (certified 28.79%, champion 29.29%). This mechanism rationalizes why external chemical interventions, such as bio-inspired antioxidants [17] (certified 29.44%, champion 29.95%) or regenerable redox shuttles [91] (champion 28.49%), provide sustained stability where structural templating alone degrades. Settling whether Sn^{2+} oxidation initiates at residual Sn-rich clusters or emerges uniformly via operando electrochemical cycling requires depth-resolved operando X-ray absorption fine-structure spectroscopy under simultaneous electrical bias and illumination, a measurement absent across these reports.

Efforts to stabilize Sn-Pb absorbers within monolithic two-terminal (2T) tandems concentrate heavily on grain-boundary and interfacial defect passivation. Modulating grain boundary energy through molecular dipoles reduces tin oxidation and defect accumulation, yielding a champion PCE of 29.67% [92], while synergistic bulk crystallization and defect management achieved 28.57% certified (champion 29.36%) [93]. Halide-cation coordination chemistries that suppress deep trap states reached a champion PCE of 29.3% [94]. Eliminating methylammonium (MA) cations mitigates volatile thermal decomposition, where sustained solvent-extraction channels in MA-free formulations delivered a certified PCE of 29.2% (champion 29.2%, 800 h T80) [13]. Molecular bridges targeting both bulk and interface defects produced a 23.3% champion PCE [95]. Recombination losses at the recombination junction and hole transport layer (hole-transport layer (HTL)) interface represent another severe bottleneck. Constructing low-acidity poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS) supramolecular complexes with self-assembled monolayers (SAMs) prevents interfacial tin etching, yielding 27.99% certified (champion 28.76%) [96]. In situ construction of PEDOT HTLs reached a 29.58% champion PCE [97]. Concurrently, solution-processed tin oxide (SnO_x) buffer layers inserted into inverted sub-cells yielded a certified PCE of 28.34% (champion 28.43%) with a 1000 h T80 [98]. Non-contact laser polishing of the perovskite surface suppressed roughness and pinholes, producing 23.47% certified (champion 29.8%) [99]. These interface modifications suppress non-radiative recombination, yet their efficacy remains restricted by the underlying chemical susceptibility of divalent tin.

Current matching requires pairing the NBG sub-cell with an efficient wide-bandgap (WBG) top cell, which introduces secondary instabilities via photo-induced phase segregation into iodide-rich and bromide-rich domains. Aligning interfacial energy levels through asymmetric SAMs with high dipole moments delivered a certified PCE of 28.98% (champion 29.35%), though measured on a miniature 0.05 cm^2 aperture [1]. Vertically oriented self-assembled monolayer (SAM) configurations achieved 19.95% certified (champion 28.59%) on an uncertified 0.0396 cm^2 device [100], whereas titanium carbide ($\text{Ti}_3\text{C}_2\text{Tx}$) MXene-SAM hybrids yielded 26.44% certified (champion 26.78%) on 0.08 cm^2 [101]. Hetero-spatial co-passivation in MA-free WBG absorbers produced a certified PCE of 20.02% (champion 28.34%) [14]. Suppressing halide segregation remains reliant on lattice-locking chemistries: halogen bonding networks that regulate bromide crystallization achieved a champion PCE of 28.58% [102], crystallographic plane pinning kinetics produced 28.24% [103], and stoichiometry-regulated crystallization delivered 29.1% [104]. Selective ion-blocking interlayers attained 28.41% champion PCE [105]. Additional passivation strategies, including bulk polymer crosslinking (champion 23.69%) [106], hydrophobic trifluoromethyl surface anchoring (champion 28.67%) [107], aromatic amino acid surface functionalization (champion 29.05%) [108], and universal composition stabilization (champion

26.75% on 0.09 cm^2) [109], mitigate surface trap densities. However, many of these approaches report champion metrics on sub- 0.1 cm^2 areas without independent certification.

Critical assessment of the 60 papers published in the all-perovskite tandem family reveals that efficiency gains remain decoupled from standardized validation. Only 25.0% of publications (15 papers) reported an independently certified efficiency, with a median certified PCE of 28.34% and a peak certified value of 30.3% [16], approaching theoretical three-dimensional finite element limits of 30.5% [110]. The remaining 75.0% relied entirely on self-reported measurements. Active area reporting is equally deficient: only 10.0% of studies stated device aperture area. The vast majority evaluated sub- 0.1 cm^2 areas, exemplified by flexible tandem modules yielding 27.5% champion PCE on 0.049 cm^2 [111]. The largest reported device area across the family reached 64.64 cm^2 through the incorporation of an antimony oxide buffer layer, which achieved a certified PCE of 27.7% (champion 28.16%) [15]. Operational stability assessments exhibit acute methodological shortcomings. Only 13.3% of papers reported a T80 value, and only 2 papers (3.3%) adhered to standardized International Summit on Organic Photovoltaic Stability (ISOS) protocols. The longest reported T80 reached 1600 h [19], but was measured in a mechanically stacked four-terminal (4T) architecture with a 21.87% champion PCE. In monolithic or encapsulated 4T tandems, reported lifetimes include 1210 h under ISOS-L-2 for a 1D/3D heterojunction (certified 21.54%) [112], 1080 h (champion 29.44%) [89], 1000 h for SnO_x -buffered cells (certified 28.34%) [98], 1000 h using a 1D perovskite interlayer (champion 28.43%) [113], 800 h for MA-free films (certified 29.2%) [13], 500 h (champion 29.18%) [88], and 300 h for 1.77 eV WBG absorbers (champion 27.26%) [114]. Moisture protection via hydrophobic capping layers tracked under ISOS-L-1 reached a 29.01% champion PCE [115], while rapid thermal cycling degraded performance to 24.31% champion PCE [116]. In contrast to single-junction perovskites (692 papers, best certified 27.6%, 63 certified, largest area 4 cm^2 , longest T80 4100 h) and hybrid perovskite/silicon tandems (77 papers, best certified 33.6%, 19 certified, largest area 16 cm^2 , longest T80 1000 h), all-perovskite tandems trail in operational endurance. Perovskite modules have demonstrated 26.8% certified PCE on 900 cm^2 with 1050 h T80 across 43 papers (10 certified). The distribution of investigative axes in this family, dominated by device architecture (34 papers) relative to defects (10 papers), interfaces (10 papers), and stability (5 papers), indicates that architectural tuning has outpaced resolution of the intrinsic chemical oxidation that limits narrow-bandgap sub-cells.

5. Hybrid Tandems: Silicon, CIGS and Organic Partners

The primary mechanistic disagreement across monolithic hybrid tandems centers on the molecular packing and structural order of the buried hole-selective layer (HSL) at the interconnection junction. One camp contends that dense intermolecular pi-pi stacking and strict crystalline order within self-assembled monolayers (SAMs) are mandatory to eliminate pinholes, establish favorable interfacial band alignment, and accelerate hole extraction, securing an independently certified power conversion efficiency (PCE) of 32.45% (33.4% champion) [117]. In direct conflict, another study demonstrates that crystalline order in hole-selective monolayers precipitates nanoscale domain boundaries, induces molecular aggregation, and accelerates thermal dewetting [23]. That group instead establishes an amorphous and asymmetric molecular geometry to suppress phase separation and maintain conformal passivation, yielding a certified PCE of 32.81% and the family-record self-reported champion PCE of 34.02% [23]. Settling whether long-range crystalline coherence or steric amorphous disorder minimizes interfacial trap-assisted recombination requires operando conductive atomic force microscopy (c-AFM) and temperature-dependent grazing-incidence wide-angle X-ray scattering (GIWAXS) conducted simultaneously under electrical bias and thermal stress to correlate nanoscale current ex-

traction with molecular reorientation. Between these extremes, intermediate chemical strategies attempt to stabilize the buried junction: sterically extended asymmetric conjugated hole-selective layers achieved a certified PCE of 33.04%, although the champion cell dropped to 24.05%, with operational durability tracked under the International Summit on Organic Photovoltaics Stability (International Summit on Organic Photovoltaic Stability (ISOS)) protocol ISOS-L-1 [118]; molecular pinning in composite SAMs on tunnel oxide passivated contact (TOPCon) silicon cells attained a certified PCE of 32.1% (32.26% champion, 1 cm^2) alongside an operational lifetime to 80% of initial performance (T80) of 1000 h [27]; siloxane-mediated intermolecular interactions in co-assembled monolayers yielded a certified PCE of 33.37% (33.74% champion) [119]; carbon-incorporated polysilicon interconnection layers achieved a certified PCE of 33.5% (33.84% champion, 1 cm^2) with an 800 h T80 [24]; four-terminal tandem antireflection and self-assembled monolayer (SAM) anchoring reached an uncertified 31.25% champion on 1 cm^2 [120]; dimeric SAMs reached 26.09% certified (26.63% champion) [121]; and dopant-free ultra-thin 2,2',7,7'-tetrakis[N,N-di(4-methoxyphenyl)amino]-9,9'-spirobifluorene (Spiro-OMeTAD) in an n-i-p architecture yielded 28.25% certified (29.73% champion) on a 0.135 cm^2 aperture [20].

Interconnection layer engineering exposes a parallel trade-off between optical parasitics, sputter damage, and series resistance. The single best certified device across the entire hybrid family reached 33.6% PCE (33.2% champion, 1 cm^2) by discarding sputtered transparent conducting oxides entirely, reporting: "For RPD-SnO_x as the recombination layer, we achieved a certified efficiency of 33.6%" [25]. By depositing reactive plasma deposition tin oxide (RPD-SnO_x) as both the recombination junction and transparent electrode, this architecture eliminates sub-bandgap absorption from indium tin oxide (ITO) and prevents high-energy ion bombardment damage to the underlying contact [25]. In contrast, sputtered ITO remains ubiquitous despite persistent energetic penalties: conformal ultrathin ITO interconnects yielded an uncertified 31.8% champion PCE [122], ultrathin electron-selective contacts yielded an uncertified 31.7% champion PCE [123], and heated ITO deposition on poly-crystalline silicon (poly-Si) passivated contacts enabled monolithic integration on a 16 cm^2 aperture, the largest physical area reported in this family, yet efficiency fell to an uncertified 22.1% champion [26]. Spatially resolved electrical characterization reveals that lateral sheet resistance across the interconnect and localized contact non-uniformities impose acute fill factor (FF) penalties, capping unoptimized monolithic output at an uncertified 29% champion PCE [124]. While scalable deposition reviews compile certified efficiencies reaching 34.85% [125], experimental monolithic scaling beyond 1 cm^2 remains bottlenecked by interfacial resistive and optical losses.

In the wide-bandgap (1.67 to 1.83 eV) perovskite top cell, open-circuit voltage (Voc) deficits caused by phase segregation and defect-mediated recombination represent the primary thermodynamic loss channel. Across the 77 papers in this family, the dominant physical mechanisms addressed are architecture (46 papers) and defects (12 papers), followed by interfaces (9 papers) and stability (4 papers). Regulating preferred crystal facets of the wide-bandgap absorber minimized the Voc deficit to deliver a certified PCE of 31.93% (32.17% champion, 1.0 cm^2), matching the median certified efficiency of the family [22]. Controlling passivator intermolecular interactions to modulate surface energetics attained a certified PCE of 33.48% (33.63% champion, 1 cm^2) [126], and targeted enhancement of passivator coordination activity enabled an open-circuit voltage of 2.00 V to reach a certified PCE of 31.52% (32.74% champion) [127]. At the buried interface, in situ engineered gradient Ruddlesden-Popper (RP) 2D/3D heterostructures suppressed non-radiative recombination to produce a certified PCE of 32.46% (32.73% champion) [21]. Uncertified champion claims addressed defect passivation and crystallographic control: bifunctional cyanoacetic acid interface-bulk

modulation achieved 32.43% [128], synchronous interfacial and crystallographic engineering yielded 32.15% [129], and grain boundary stabilization reported a 32.19% champion despite certifying at only 23.48% [130]. In parallel, theoretical bromine and cesium substitution in formamidinium lead iodide (FAPbI₃) calculated a 32.1% champion potential for perovskite/TOPCon tandems [131], surface reconstruction through phenylethylammonium chloride (PEACl) modification and solvent etching reached 31.85% [132], imide passivation achieved 31.66% [133], volatile fluorination additives yielded 31.43% [134], pathway priming of intermediate phases in formamidinium cesium (FACs) absorbers reached 31.04% on 1.0 cm² [135], simulated low-deficit wide-bandgap tandems claimed 30.97% against a certified baseline of 23.42% [136], electron-deficient amine hydrogen bonding yielded 26.76% [137], and crystallization modulation with defect passivation achieved 20.4% [138].

Deposition pathways across the six-month window increasingly diversified to eliminate solvent-induced degradation on textured bottom substrates. Bandgap-tunable two-step vapor deposition produced monolithic perovskite/silicon tandems with an uncertified 29% champion PCE [139]. Solvent-free close-space sublimation (CSS) demonstrated conformal coverage over textured silicon wafers, achieving an uncertified 24.3% champion PCE [140]. Rapid laser-based deposition of lead iodide and cesium bromide (PbI₂/CsBr) inorganic scaffolds enabled hybrid fabrication across 1 cm² [141]. Expanding multijunction architectures, chloride substitution enabled thick perovskite middle absorbers in monolithic perovskite/perovskite/silicon triple-junction cells evaluated on 1 cm² [142].

Hybrid tandems also incorporate non-silicon bottom partners, specifically organic bulk heterojunctions (BHJs), copper indium gallium selenide (CIGS), cadmium telluride (CdTe), and lead sulfide (PbS) colloidal quantum dots. These alternatives target mechanical flexibility, lightweight form factors, high-throughput roll-to-roll processing, and infrared photon harvesting. However, these benefits come at substantial physical cost: severe photovoltage deficits, non-radiative recombination at rough chalcogenide interfaces, narrow thermal stability windows in organic acceptors, and large conduction band offsets. In perovskite/organic tandems, localized 2D/3D heterojunctions enhanced photovoltage to yield a certified PCE of 26.3% (27.11% champion) [143], absorption tuning and energy loss minimization achieved 25.56% certified (26.49% champion) [144], blade-coated wide-bandgap perovskites addressing halide ion homogeneity reached 25.6% certified (26.3% champion) [145], and aggregation-induced emission luminogens incorporated into ternary organic bulk heterojunctions yielded a certified PCE of 25.8% (26.5% champion) [146]. Uncertified configurations exhibited matching ceilings: synergistic hole-selective contacts achieved 26.06% [147], gradient-regulated crystallization reached 25.91% [148], and defect passivation of 1.83 eV perovskites on flexible substrates yielded 22.44% [149]. For chalcopyrite partners, coevaporated monolithic perovskite/CIGS tandems reached a certified PCE of 24.5% on 1 cm² [150], while oriented crystal growth on flexible CIGS achieved an uncertified 25.02% champion PCE [151]. Four-terminal tandems coupling halide-tuned cesium lead mixed halide (CsPbX₃) top cells with CdTe bottom partners achieved an uncertified 25.13% champion PCE [152], and sulfur-supplemented PbS quantum dot cells reached an uncertified 14.34% champion PCE [153].

Weighing these metrics against other photovoltaic classes reveals severe reporting discrepancies across the hybrid tandem family. Of the 77 papers published in this family over six months, only 24.7% (19 papers) reported an independently certified efficiency, with 11 papers excluded from empirical tallies as simulations or reviews. Active device area was documented in only 15.6% of publications, reaching a maximum of 16 cm² [26], while champion devices remain overwhelmingly restricted to 1 cm² or sub-0.15 cm² apertures. Operational durability reporting is exceptionally sparse: only 2.6% of papers reported a T80 metric, with the longest reaching 1000 h [27], and only a single paper (1.3%) specified

a standard testing protocol (ISOS-L-1) [118]. By comparison, single-junction devices (692 papers) demonstrated certified efficiencies up to 27.6% (63 certified) with operational stability reaching 4100 h on 4 cm^2 ; all-perovskite tandems (60 papers) reached 30.3% certified (15 certified) across areas up to 64.64 cm^2 and 1600 h T80; and perovskite modules (43 papers) delivered 26.8% certified (10 certified) across 900 cm^2 with 1050 h T80. Although perovskite/silicon and hybrid tandems command the global photovoltaic efficiency frontier at 33.6% certified, their near-complete absence of standardized long-term stability testing and severe efficiency decline upon scaling past 1 cm^2 present an urgent barrier that laboratory efficiency claims currently obscure.

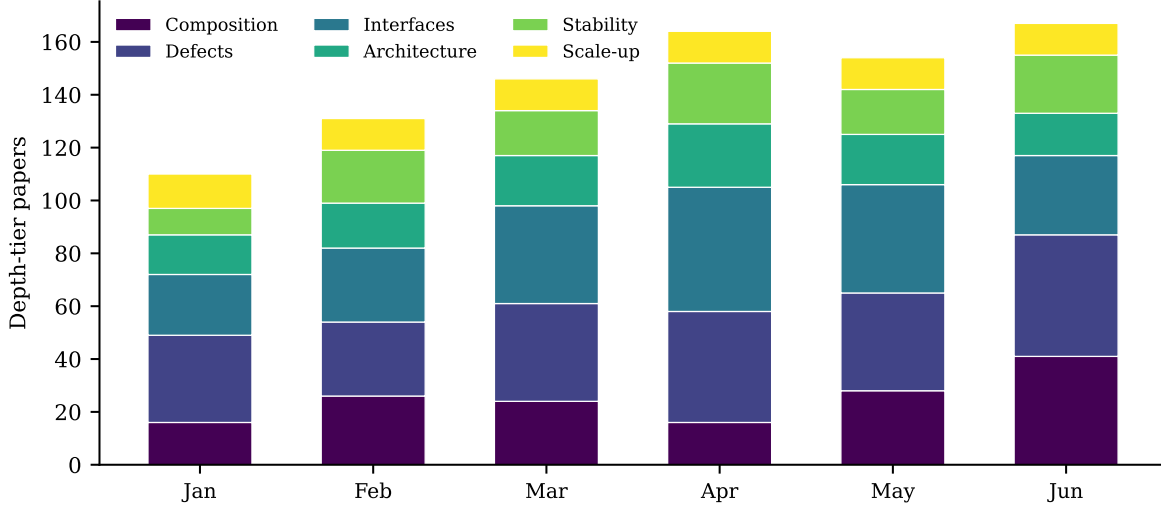


Figure 3. Where the month’s mechanistic effort was directed. Depth-tier papers by mechanism axis, stacked by device family.

6. Modules and the Cost of Area

Recent module research exposes a direct contradiction regarding buried contact chemistry: whether large-area crystallization requires intentional metal excess to direct unidirectional growth, or strict stoichiometric neutrality to eliminate trap-assisted recombination. One approach demonstrates that an engineered lead-rich (Pb-rich) buried interface enforces upward unidirectional crystallization from the substrate, eliminating buried voids and delivering an uncertified 21.56% champion power conversion efficiency (PCE) across a 10.08 cm^2 carbon-electrode perovskite solar module (PSM) [154]. Conversely, area-selective atomic layer deposition (ALD) studies demonstrate that undercoordinated interfacial lead atoms act as non-radiative recombination centers rather than benign growth directors, requiring passivation with aluminium oxide (AlOx) to achieve an uncertified 26.41% champion PCE across 64.68 cm^2 [155]. This divergence extends across alternative contact strategies: mesoscopic chemical reconfiguration of the tin oxide (SnO_2) electron transport layer achieved a certified 26.64% PCE (26.7% champion) on 21.5 cm^2 [31], while redox-mediated buried interface stabilization yielded an uncertified 26.48% champion PCE on 11.0 cm^2 [156]. Resolving whether residual interfacial lead directs crystallization or accelerates phase segregation requires operando, cross-sectional photoluminescence (PL) lifetime imaging coupled with nanoscale secondary ion mass spectrometry (nanoSIMS) across module stripe widths under continuous bias.

Scalable meniscus coating reveals an analogous conflict in fluid mechanics: whether large-area unifor-

mity requires prolonged drying to suppress Marangoni flows or rapid extraction to bypass intermediate solvate phases. Suppressing coffee-ring flows through fluid-locking interfacial layers yielded an uncertified 26.61% champion PCE [157], while pairing fluid dynamics with ambient-printed self-assembled monolayer (SAM) interlayers achieved a certified 25.85% PCE (26.31% champion) across 13.04 cm² [158]. Conversely, slot-die processing of formamidinium lead iodide (FAPbI₃) across 59.5 cm² required forced solvent extraction to suppress yellow-phase formation, attaining an uncertified 20.8% champion PCE [159]. Solvent formulation deepens this contrast: localized solidification control with green colloidal inks produced an uncertified 22.3% champion PCE across 31.5 cm² [160], whereas chemical soaking baths designed to rapidly quench wide-bandgap films produced an uncertified 22.03% champion PCE over 10.13 cm² [161], matching frameworks for roll-to-roll slot-die coating [162]. These contradictory processing windows show that scalable printing lacks a unified model relating ink rheology, evaporation rate, and nucleation kinetics.

Molecular design of hole-selective self-assembled monolayers (SAMs) targets series resistance across extended substrates, yet consensus is lacking on whether molecular rigidity or mechanical compliance governs scalability. Imposing rotational constraints on phosphonic acid anchors aligns interfacial dipoles and minimizes potential fluctuations, yielding a certified 26.31% PCE (26.87% champion) across a 47.84 cm² aperture [163]. Asymmetric polymeric SAMs homogenize buried energetics to deliver an uncertified 26.13% champion PCE on 57.6 cm² [33], and ionic SAMs establish neutral interfaces that produce an uncertified 26.88% champion PCE across 29.7 cm² [164]. However, rigid dipolar ordering often sacrifices interfacial mechanical toughness: incorporating benzofuran-substituted molecules improves adhesion but caps the uncertified champion PCE at 21.2% on 10.0 cm² [165], while bilateral anchoring strategies attempt to balance charge transfer and adhesion [166]. The persistent efficiency roll-off with area indicates that wetting heterogeneity and pinholes through the SAM layer dominate scaling losses over intrinsic dipole alignment.

Thermomechanical stress represents an escalating barrier to module longevity, as lattice expansion mismatches accelerate defect formation. Homogenizing the out-of-plane strain distribution across the absorber yielded a certified 25.55% PCE (26.59% champion) across a 100 cm² flexible module under International Summit on Organic Photovoltaics Stability (International Summit on Organic Photovoltaic Stability (ISOS)) ISOS-L-1 illumination [30]. Regulating crystallization-defect-strain interactions in formamidinium-cesium (FACs) alloys produced a certified 26.12% PCE (26.76% champion) across 12 cm² [167]. Cross-linking scaffolds reinforce grain boundaries against strain-induced degradation: a sodium-integrated cross-linkable scaffold enabled a certified 26.75% PCE (26.9% champion) on 10.24 cm² [29], while cross-linked n+-type interlayers achieved a certified 26.27% PCE (26.34% champion) over 14 cm² [168]. Fluorinated self-polymerizable additives passivate halide vacancies for an uncertified 26.01% champion PCE on 14 cm² [169], and conjugated imidazolium additives inhibit ion migration to achieve an uncertified 25.63% champion PCE on 20.25 cm² [170]. Vapor-phase grain-boundary anchoring provides bending endurance on flexible substrates, though uncertified champion PCE remains at 15.7% across 20.25 cm² [171].

Across the 43 papers in this module family, the area penalty remains severe: while single-junction small cells reach 27.6% certified PCE on areas up to 4 cm², modules exhibit a median certified PCE of 26.27%, with only 23.3% of studies reporting independent certification and 69.8% stating an active area. The highest certified module efficiency was 26.8% (27.12% uncertified champion) using a self-healing dynamic-bond scaffold across 655.2 cm² [32]. Dual-targeted anchoring delivered an uncertified 26.62% champion PCE across 651 cm² under verified ISOS-L-1 testing [172], and in-situ

dual-stage reactions sustained operation at an uncertified 23.3% champion PCE over 694.5 cm² [173]. Modular auxetic metamaterials integrating silver nanowire (AgNW) and graphene networks reached 900 cm², the largest area in the family [34]. Scribe-line dead zones and thermal edge degradation constitute intrinsic area costs: controlling crystallization kinetics at laser scribe lines achieved a certified 23.55% PCE (24.7% champion) on 100 cm² under ISOS-D-1 dark storage [174]. Scribing and geometry adjustments also support vapour-treated printed carbon modules [175] and optical front layers for bifacial modules delivering an 18.48% certified PCE [176].

Low-dimensional heterojunctions and composition pinning provide chemical stabilization, yet standardized stability reporting remains rare: only 4.7% of module papers reported a time to 80% of initial performance (T80), and only 7.0% (3 papers) followed defined ISOS protocols. Intercalated 2D/3D perovskite heterojunctions achieved an uncertified 26.22% champion PCE across 6.8 cm² with a T80 of 1050 h, the longest lifetime in this family [35]. Non-Dion-Jacobson 2D capping layers achieved an uncertified 26.52% champion PCE over 50 cm² [177], thermotropic liquid interface regulation yielded a certified 25.62% PCE on 22.8 cm² [28], and composition pinning in lead-tin perovskites delivered an uncertified 19.37% champion PCE across 64 cm² [178]. Semi-transparent cesium lead bromide (CsPbBr₃) modules reached an uncertified 7.78% champion PCE across 64 cm² [179]. Tandem module implementations indicate that scaling penalties persist across multi-junction geometries relative to 16 cm² hybrid tandems (33.6% certified) and 64.64 cm² all-perovskite tandems (30.3% certified): four-terminal perovskite/silicon tandem modules with MXene junctions achieved an uncertified 19.45% champion PCE across 60 cm² [180], and perovskite-organic tandems reached an uncertified 19.89% champion PCE across 16.41 cm² [181]. Finally, indoor photovoltaics (IPVs) function under narrow low-flux spectra and must not be benchmarked against one-sun metrics: fully evaporated mini-modules reached an uncertified 41.6% champion PCE under indoor lighting, specifically 41.60% at 900 lux and 41.22% at 300 lux across 3.9 cm² [81], while semi-transparent indoor modules delivered an uncertified 13.78% champion PCE with a 268 h T80 [182].

Table 3. Reported efficiency against device area

Aperture area (cm ²)	Devices	Best certified	Best self-reported	Median self-reported
< 0.1	33	28.98%	29.35%	26.47%
0.1 - 1	11	28.25%	29.73%	22.51%
1 - 10	24	33.6%	33.84%	26.22%
10 - 100	26	32.95%	28.16%	26.01%
> 100	5	26.8%	27.12%	26.62%

Every device in the half year that reported an aperture or active area, binned by decade. The median column is the more informative one: a best value in a band can be a single outlier, while the median describes what the band routinely achieves. Devices reporting no area cannot appear here, which is itself a reporting limitation rather than an absence of large-area work.

7. Reading the Device Families Together

A fundamental disagreement across device architectures centers on whether charge extraction through buried hole-selective contacts requires strict crystalline molecular order or steric amorphous disorder. In monolithic hybrid tandems, one investigation asserts that dense intermolecular pi-pi stacking

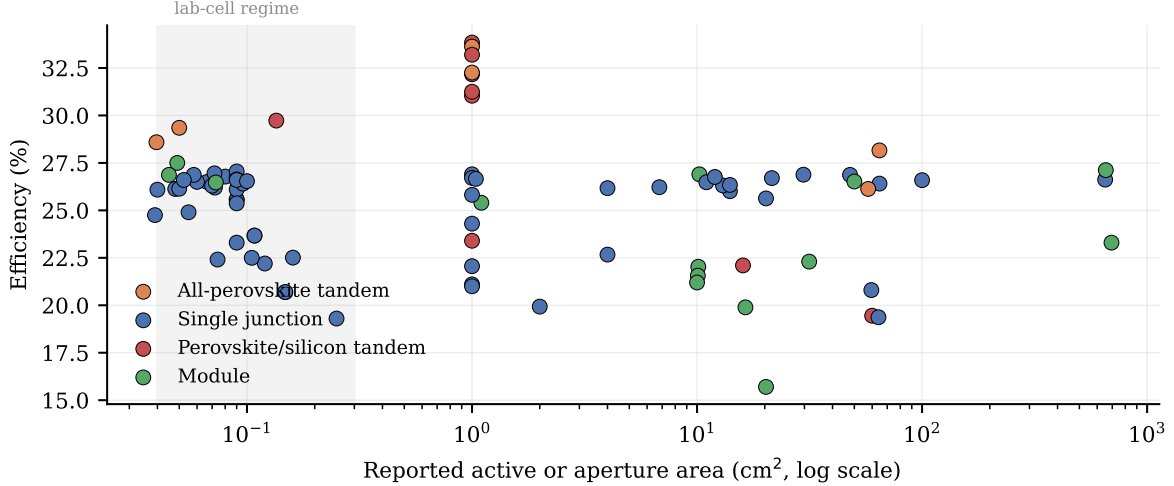


Figure 4. Efficiency against reported active or aperture area. The shaded band marks the sub-0.3 cm² laboratory-cell regime in which most champion values are measured.

and rigid crystalline order within self-assembled monolayers (SAMs) are essential to eliminate pin-holes, align bands, and accelerate extraction, securing an independently certified power conversion efficiency (PCE) of 32.45% (33.4% champion) [117]. In direct conflict, another study demonstrates that crystalline order in hole-selective monolayers precipitates nanoscale domain boundaries, induces aggregation, and accelerates thermal dewetting, requiring an amorphous, asymmetric molecular geometry to suppress phase separation and preserve conformal passivation (certified PCE of 32.81%, family-record 34.02% champion) [23]. In single junctions, this contact is interpreted instead through electrostatic dipole models where work-function shifts govern extraction (certified PCE of 26.23%, champion 26.18%) [47], whereas competing reports attribute open-circuit voltage (V_{oc}) gains to chemical termination of undercoordinated ions and hydroxyl defects on nickel oxide (NiO_x) (certified PCE of 26.2% on 0.0524 cm²) [42,50]. Settling whether extraction is governed by long-range crystalline order, steric amorphous disorder, or chemical hydroxyl passivation requires operando conductive atomic force microscopy (c-AFM) and temperature-dependent grazing-incidence wide-angle X-ray scattering (GIWAXS) conducted simultaneously under operational electrical bias and thermal stress to correlate nanoscale current percolation directly with molecular reorientation.

Reading the four device families together reveals that identical physical failures recur across architectures under divergent terminology. In single-junction cells, absorber decay is framed as lattice microstrain and octahedral distortion, where static molecular confinement (certified PCE of 26.79% under International Summit on Organic Photovoltaic Stability (ISOS) protocol ISOS-L-1) [61] conflicts with dynamic photoswitchable isomers that dissipate cyclical thermal fatigue (certified PCE of 26.9%, champion 27.2%) [3] or dual-dynamic covalent seeding (certified PCE of 26.69%, protocol ISOS-L-2I) [53]. In modules, this mechanochemical instability resurfaces as convective drying gradients and heterogeneous out-of-plane residual lattice strain across macroscopic substrates. Homogenizing out-of-plane strain enabled an independently certified PCE of 25.55% (champion 26.59%) across a 100 cm² flexible module under protocol ISOS-L-1 [30], while fluid-locking interlayers suppressed convective solute segregation to reach an uncertified 26.61% champion PCE [157] and ambient fluid control achieved a certified PCE of 25.85% on 13.04 cm² [158]. Interfacial module loss is concurrently assigned to laser scribe margins, where regulating crystallization kinetics yielded a certified PCE of

23.55% across 100 cm^2 under protocol ISOS-D-1 [174]. Thus, what single-junction studies isolate as microscopic octahedral distortion drives the macroscopic drying fissures, scribe recombination, and strain gradients that penalize large modules.

Conversely, chemical interventions validated in one device family introduce severe transport penalties when transferred into another. In single-junction devices, depositing wide-bandgap two-dimensional (2D) or quasi-2D caps over three-dimensional (3D) absorbers suppresses non-radiative recombination and moisture ingress, yielding a certified PCE of 27.02% under protocol ISOS-L-2I [36], a certified PCE of 26.2% via in situ guanidinium heterostructures [38], and a certified PCE of 21.69% in quasi-2D films [77]. In modules, surface 2D/3D capping provides environmental defense, reaching a 26.22% champion PCE across 6.8 cm^2 with a 1050 h time to 80% of initial efficiency (T80) [35] and a 26.52% champion PCE on 50 cm^2 using non-Dion-Jacobson diammonium cations [177]. However, when transferred into monolithic two-terminal (2T) tandems, these low-dimensional caps introduce anisotropic out-of-plane transport barriers from disordered organic spacers. Although single-junction cells tolerate minor extraction resistance, series-connected tandems suffer acute fill factor (FF) losses and current mismatch under illumination. Single-junction and module devices avoided insulating phases through vapor-mediated reconstruction enforcing vertical transport (certified PCE of 26.15% on 1 cm^2) [40] or multivalent oriented ligands (certified PCE of 25.4% on 1.1 cm^2) [37], whereas hybrid tandems required gradient Ruddlesden-Popper heterostructures to contract current mismatch to 0.1% and reach a certified PCE of 32.46% [21].

The four families also impose incompatible demands on the same functional layers. At the interconnection junction of hybrid tandems, minimizing optical parasitics favors discarding sputtered transparent conducting oxide (TCO) layers, demonstrated by reactive plasma deposition tin oxide (RPD-SnOx) achieving the record certified PCE of 33.6% (champion 33.2% on 1 cm^2) [25]. Yet omitting lateral conductivity is unviable in modules and monolithic tandems, where current non-uniformities across textured pyramids or scribe lines create localized resistive bottlenecks, capping unoptimized monolithic output at an uncertified 29% champion PCE [124], while sputtered indium tin oxide (ITO) on 16 cm^2 drops hybrid tandem output to an uncertified 22.1% champion [26]. In addition, monolithic integration converts current mismatch into an electrochemical destruction route: in all-perovskite tandems, current mismatch drives the narrow-bandgap (NBG) mixed tin-lead (Sn-Pb) subcell into reverse bias, where hole injection triggers an autonomous redox cascade between divalent tin (Sn^{2+}) and oxidized halides that degrades the lattice (champion 29.03%) [90], accompanied by cyclic iodine redox at buried contacts (certified PCE of 28.79%, champion 29.29%) [18]. Synchronizing crystallization kinetics attains a certified PCE of 30.3% [16], but cannot prevent this reverse-bias decay.

Read together, these four device families do not share a single optimization pathway. Single junctions operate in an electrostatic regime on sub- 0.1 cm^2 areas; hybrid tandems are governed by optical parasitics and buried contact order; all-perovskite tandems are constrained by reverse-bias redox degradation; and modules remain bottlenecked by convective drying instabilities, scribe series resistance, and mechanical shear across expanded apertures up to 900 cm^2 [34]. Treating them as sequential milestones of a single technology obscures the reality that each family occupies a distinct physical problem space.

8. How Each Device Family Reports Itself

A fundamental disagreement in inverted perovskite solar cells concerns how self-assembled monolayer (SAM) hole-selective contacts degrade during operation: while one study attributes operational failure to out-of-plane molecular desorption from metal oxides that rigid planar conjugation suppresses [183], another concludes that degradation proceeds via field-induced lateral aggregation into isolated clusters [7]. Resolving whether molecular detachment or lateral dewetting governs contact failure requires operando bias-dependent sum-frequency generation spectroscopy paired with in situ conductive atomic force microscopy under continuous maximum power point tracking (MPPT) at operational temperatures. Reconciling conflicting degradation mechanisms across the literature is hindered by scarce standardized testing: when stability assessments use disparate stress regimes, distinguishing true stabilization from divergent testing conditions becomes impossible.

Comparing reporting practices across device categories demonstrates that the families do not verify themselves to the same standard. These detection rates reflect automated parsing of published abstracts rather than exhaustive laboratory audits; omitting a testing protocol or active area in an abstract does not establish that researchers failed to execute the measurement. Across the 2238 indexed abstracts, 52.1% stated a power conversion efficiency (PCE) and 8.5% reported independent certification. However, these pooled averages conceal stark disparities among device families. Single-junction devices encompass 692 papers, yet only 53.0% stated an efficiency and 9.1% reported independent certification. Single junctions dominate volume but exhibit the lowest certification rate, relying on unverified laboratory scans susceptible to capacitive transients and spectral mismatch. In contrast, multi-junction and module families face higher verification demands to justify architectural complexity: all-perovskite tandems (60 papers) stated efficiency in 71.7% and independently certified 25.0%, perovskite/silicon and other hybrid tandems (77 papers) stated efficiency in 64.9% and certified 24.7%, and modules (43 papers) stated efficiency in 72.1% and certified 23.3%. Certification rates in tandems and modules are nearly triple that of single junctions, establishing a divided evidentiary standard.

Geometric reporting similarly divides module architectures from cell-level studies. Modules report active area in 72.1% of papers, compared to only 6.6% for single junctions, 10.0% for all-perovskite tandems, and 15.6% for hybrid tandems. Because module performance is governed by geometric dead area and interconnect resistance, area disclosure is unavoidable; in contrast, single-junction literature routinely omits aperture size, obscuring whether efficiencies reflect scalable areas or microscopic crystals. Operational durability tracking reveals further divergent priorities: all-perovskite tandems lead time to 80% of initial performance (T80) reporting at 13.3%, compared to 3.9% for single junctions, 4.7% for modules, and 2.6% for hybrid tandems, reflecting scrutiny of narrow-bandgap mixed tin-lead subcells vulnerable to tin oxidation. However, standardized degradation protocols remain universally rare: an International Summit on Organic Photovoltaic Stability (ISOS) protocol was specified in only 3.6% of single-junction, 3.3% of all-perovskite tandem, 1.3% of hybrid tandem, and 7.0% of module abstracts. Hybrid tandems exhibit the sharpest disparity: despite achieving 24.7% certification, only 1.3% specified an ISOS protocol and 2.6% reported a T80 lifetime, substituting unstandardized illumination for standardized aging.

Exemplary reporting occurs where geometry, certification, and operational protocols converge. Modules establish reproducible baselines by reporting aperture dimensions, certified PCE, and standardized degradation conditions, including an active area of 100 cm^2 with a certified PCE of 25.55% under ISOS-L-1 tracking [30], 100 cm^2 with a certified PCE of 23.55% under ISOS-D-1 storage [174], or a

certified 26.8% on a 655.2 cm² module [32]. Standardized tandem benchmarks report certified efficiencies on designated apertures, such as certified 33.6% on 1 cm² [25] and certified 32.1% on 1 cm² with a T80 of 1000 h [27] in hybrid tandems, or certified 27.7% on 64.64 cm² [15] and a T80 of 1600 h [19] in all-perovskite tandems. Rigorous single-junction reporting achieves certified 27.6% PCE [8], certified 26.78% under ISOS-L2 [6], or a T80 of 4100 h [12]. In sharp contrast, absent reporting leaves claims unanchored. Uncertified single-junction hero cells reach 28.49% PCE while omitting certification, active area, and standardized aging [91]. Uncertified records remain confined to sub-0.1 cm² cells, including 0.072 cm² at 26.2% PCE [184] and 0.058 cm² at 26.87% PCE [43], where edge collection distorts planar transport [43,184]. Conversely, module studies report a 900 cm² aperture without independent certification [34]. Without uniform reporting standards across device families, comparing headline efficiencies mistakes divergent verification rigor for genuine physical progress.

Table 4. Reporting completeness by device family

Device family	Papers	States an efficiency	Independently certified	States an area	Reports T80	Names an ISOS protocol
Single junction	692	52.5%	9.1%	6.6%	3.9%	3.6%
All-perovskite tandem	60	71.7%	25%	10%	13.3%	3.3%
Hybrid tandem	77	63.6%	24.7%	15.6%	2.6%	1.3%
Module	43	69.8%	23.3%	69.8%	4.7%	7%

Percentages are of that family’s depth-tier papers. These are detection rates over abstracts, not audits of experimental practice: a protocol that is not named in an abstract may still have been followed, so each figure is a lower bound. The ISOS column counts a specified protocol such as ISOS-L-1; a paper referring to ISOS protocols without naming one is not counted.

9. Research Gaps and Outlook

Incompatible conclusions govern buried hole-selective contacts across inverted and tandem architectures, dividing between crystalline molecular order and steric amorphous disorder. In monolithic hybrid tandems, dense pi-pi stacking and crystalline order within self-assembled monolayers (SAMs) are argued to eliminate pinholes and accelerate hole extraction, securing a certified power conversion efficiency (PCE) of 32.45% (33.4% champion) [117]. Conversely, competing evidence shows that crystalline order precipitates domain boundaries and accelerates thermal dewetting, requiring an amorphous, asymmetric molecular geometry to suppress phase separation (certified PCE of 32.81%, family-record 34.02% champion) [23]. Failure pathways also conflict: while one study attributes operational decay to out-of-plane molecular desorption from metal oxides [183], another shows failure proceeding via field-induced lateral aggregation into clusters [7]. The half year’s evidence does not establish whether buried contacts require crystalline coherence or steric disorder, nor whether failure initiates by chemical detachment or lateral dewetting. Resolving this controversy requires operando bias-dependent sum-frequency generation spectroscopy paired with temperature-dependent grazing-

incidence wide-angle X-ray scattering (GIWAXS) and in situ conductive atomic force microscopy (c-AFM) under continuous maximum power point tracking (MPPT) at elevated temperatures.

A second unresolved gap concerns low-dimensional capping layers in single-junction perovskite solar cells (PSCs). Wide-bandgap two-dimensional (2D) or quasi-2D caps over three-dimensional absorbers are reported to suppress non-radiative recombination without degrading collection, yielding a certified PCE of 27.02% (27.21% champion) under International Summit on Organic Photovoltaic Stability (ISOS) protocol ISOS-L-2I [36], an in situ 2D/3D heterostructure delivering 26.2% certified PCE (26.49% champion) [38], and quasi-2D passivation reaching 21.69% certified PCE (22.76% champion) [77]. Opposing studies show that solution-processed 2D caps deposit disordered, insulating spacers with random quantum-well orientations, penalizing fill factor (FF) under illumination and requiring vapor reconstruction (certified PCE of 26.15%, 26.71% champion on 1 cm^2) [40] or multivalent directing agents (certified PCE of 25.4%, 25.4% champion on 1.1 cm^2) [37]. Because steady-state photoluminescence cannot distinguish defect passivation from resistive charge accumulation, determining whether 2D caps act as extraction conduits or resistive dead zones requires operando depth-resolved transient absorption spectroscopy paired with cross-sectional c-AFM under operational bias and 1-sun illumination.

A third gap specific to all-perovskite tandems divides over whether narrow-bandgap (NBG) mixed tin-lead (Sn-Pb) degradation is an extrinsic crystallization artifact or an intrinsic electrochemical consequence of operating redox cascades. One view attributes Sn-Pb decay to differential crystallization where tin iodide precipitates prematurely; synchronizing crystallization via chemical hardness engineering achieved a certified PCE of 30.3% (champion 30.3%) [16], while suppressing Sn-rich precipitation yielded a certified PCE of 28.53% (28.77% champion) [87]. Conversely, competing reports demonstrate that even in uniform absorbers, reverse bias triggers redox cascades between Sn^{2+} and oxidized halides (champion 29.03%) [90], while cyclic iodine reactions regenerate triiodide that oxidizes Sn^{2+} to Sn^{4+} (certified PCE of 28.79%, 29.29% champion) [18]. Current data do not establish whether crystallographic control prevents Sn^{2+} oxidation under bias. Settling whether oxidation initiates at localized Sn-rich phases or propagates electrochemically requires operando depth-resolved X-ray absorption fine structure (XAFS) spectroscopy coupled with in situ electrochemical impedance spectroscopy under operational bias and illumination.

A fourth gap centers on the efficiency penalty when scaling perovskite solar cells into perovskite solar modules (PSMs). Evidence divides on whether area-dependent deficits stem from laser scribe margins, convective drying instabilities, or bulk lattice strain. Regulating crystallization at laser scribes delivered a certified PCE of 23.55% (24.7% champion) across 100 cm^2 under ISOS-D-1 dark storage [174], identifying scribe damage as the primary loss pathway. Conversely, competing models show that convective drying instabilities and coffee-ring segregation dominate scaling deficits, where fluid-locking interlayers reach an uncertified 26.61% champion PCE [157] and ambient fluid-crystallization control yields a certified PCE of 25.85% (26.31% champion) on 13.04 cm^2 [158]. Contradicting both, other work attributes deficits to residual strain, where strain homogenization achieved a certified PCE of 25.55% (26.59% champion) on 100 cm^2 under ISOS-L-1 [30] and strain coordination reached a certified PCE of 26.12% (26.76% champion) on 12 cm^2 [167]. Isolating geometric interconnect losses from hydrodynamic drying non-uniformities requires calibrated spatial photoluminescence quantum yield imaging synchronized with micro-beam X-ray diffraction across active strips and scribe lines under bias.

Underpinning all four device families is an acute deficit in standardized operational durability data.

Across 872 closely read papers in the six-month corpus, only 39 reported a time to 80% of initial efficiency (T80) lifetime, longest 4100 h, and only 39 named any stability protocol. Independent certification remains sparse: among 692 single-junction papers (best certified 27.6%, largest area 4 cm², longest T80 4100 h), only 63 reported certified values; among 60 all-perovskite tandem papers (best certified 30.3%, largest area 64.64 cm², longest T80 1600 h), only 15 were certified; among 77 hybrid tandem papers (best certified 33.6%, largest area 16 cm², longest T80 1000 h), only 19 were certified; and among 43 module papers (best certified 26.8%, largest area 900 cm², longest T80 1050 h), only 10 were certified. Because stability assessments use disparate stress regimes, this half year’s evidence could not support an equitable cross-family comparison of operational lifetime between single-junction cells and monolithic hybrid tandems, where higher tandem efficiencies cannot be evaluated against single-junction durability. To make device categories comparative, the field must report accredited steady-state power conversion efficiency on verified apertures alongside continuous maximum power point tracking under standardized ISOS protocols (specifically ISOS-L-2 and ISOS-L-3 at 65 °C or 85 °C) with mandatory disclosure of spectral mismatch factors and fill factor loss mechanisms.

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Declaration of Competing Interest

The authors declare no conflict of interest.

Data Availability Statement

The corpus table for the month, the per-paper extraction records with their verbatim source quotations, the statistics file underlying every number in the text, and the gate report are provided as Supplementary Information. Abstract text and full texts are not redistributed.

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